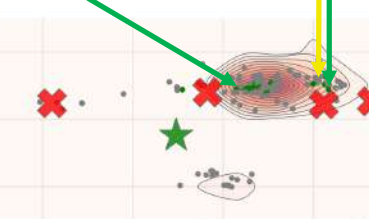


Wait, that seems high. **Let me check** my calculations again. Maybe I made a mistake somewhere.

Wait, that might be the correct way to model it. Let me denote

Wait, is that right? Let me double-check. $420 \text{ units/month} \times 12 \text{ months} = 5,040 \text{ units}$. $5,040 \times \$110 = \$554,400$. Yeah, that seems correct



0-20% states

But **wait**, the initial number of marbles isn't given, so the total number in the bag must be unknown.

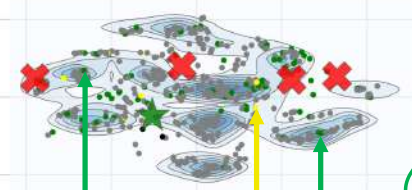
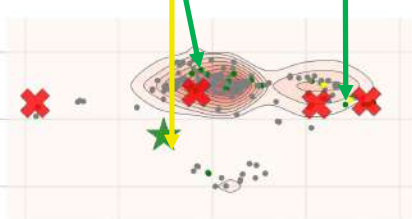
Wait, the question is asking for the greatest possible value for the probability that neither happens.

Let me check the options. I don't think I made any mistakes here because the steps align with the definition of independent events in probability.

Wait, but that can't be right because the selling price has to cover the costs. Wait, **let me check** my calculations again because I think I might have made a mistake somewhere.

Wait, that seems a bit messy. Maybe there's another way.

Wait, but let me double-check my calculations to make sure I didn't make a mistake.



20-40% states

Wait a second, but initially, after adding the 20 black marbles, the number of black marbles is 20.

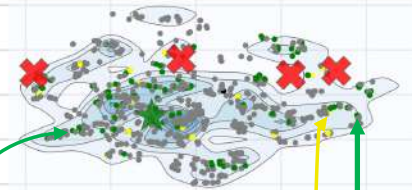
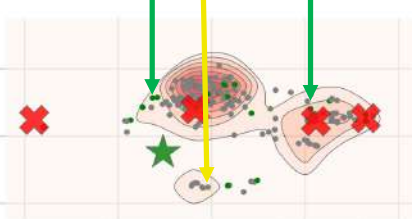
Wait, but I should double-check that. Yes, $\tan(60^\circ) = \sqrt{3}$, so the adjacent side (distance to the base) is $h / \sqrt{3}$ when the angle is 60° .

Wait, **let me check**: 1.2 times what gives 360. Maybe I can do this step by step. 1.2 is the same as $6/5$, so dividing by 1.2 is the same as multiplying by $5/6$.

Wait, **let me check** because I think I might have made a miscalculation. Let me try another way

Wait, maybe another way: ...

Wait, perhaps I should compute the per unit numbers differently. Let me recalculate.



40-60% states

Wait, but let me double-check that. If the sales-person makes 40 sales, each at \$250, their commission would be 40 times \$25, which is \$1000 exactly.

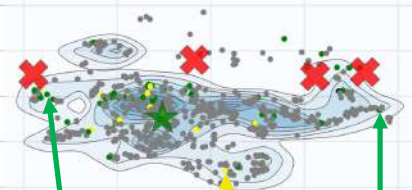
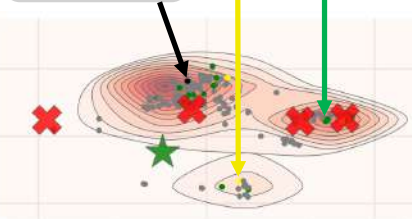
Wait, but let me double-check in case I made a mistake in simplifying.

Hmm, so the additional time required is $5(\sqrt{3} + 1)$ minutes. **Let me check** the options given.

But just to be absolutely sure, **let me check** with another option, say option A is \$61.

But this seems a bit confusing. Maybe I **made a mistake** in the algebra.

But **wait**, maybe I made an error earlier. Let me go back and check my steps again.



60-80% states

But **wait**, the options are A)21, B)30, C)31, D)32, E)None of the above. So, D is available. Therefore, the average is 32, which matches option

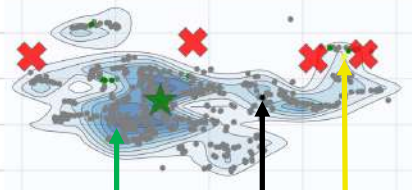
But **wait**, sometimes when I do these problems, I might have missed something. Let me think again.

Wait, but **let me check** again because sometimes rounding can cause issues. Let me recalculate 21.90 divided by 0.28 more accurately.

Wait, but **let me check** if $S=230$ would be sufficient.

Wait, that seems similar to what I had before.

But **wait**, let me double-check my initial calculation because 78.20 is an option, but perhaps I made a mistake in my first equation setup.



80-100% states

Wait, no, in the initial example, it's a different problem. So I think my logic here is correct.

But let me make sure I didn't **make a mistake**... Yes, that seems correct.

But wait, **let me check** if 234.2857 would satisfy the condition. Let's plug it back in.

● thought ● let me check ● wait ● make a mistake